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Generalized Multi-Output High-Gain Observer with Application to Ego Vehicle Trajectory and Orientation Estimation

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Abstract

This study addresses the critical issue of vehicle localization, using a novel enhanced multi- output nonlinear high-gain observer. The nonlinear observer is specifically designed to estimate the real-time trajectory and orientation of the ego vehicle, which are needed in steering control and other autonomous driving applications. Unlike a traditional high-gain observer, the new nonlinear observer method proposed in this paper ensures input-to-state stability, is applicable to systems in a more general system form, and can accommodate additional measurements as well as constraints in the system dynamics. The paper first outlines the new high-gain observer design methodology for a dynamic system with multiple outputs. The performance of the observer is then evaluated through simulations and real-world experimental tests using the KITTI dataset. Both types of results confirm the effectiveness and robustness of the observer in tracking the ego vehicle trajectory and estimating vehicle states.

Key words: Observer design; High-gain observer ; LPV/LMI observer ; Autonomous Vehicles; Ego-vehicle estimation.

1 Introduction

The precise and robust estimation of vehicle variables plays a pivotal role in a wide range of vehicular technologies (Kumar *et al.*, 2013; Wang and Sun, 2023), from driver assistance systems to autonomous driving. However, achieving accurate and reliable estimation presents a multitude of challenges that require sophisticated solutions. Vehicle systems are inherently complex and nonlinear, with dynamics influenced by various external factors like sensor noise, actuator faults, and environmental disturbances. These uncertainties can significantly disrupt estimation accuracy, making it difficult to predict and reconstruct the true state of the vehicle. While sensors provide valuable measurements, they inherently offer incomplete information about the vehicle state (Wang and Sun, 2023).

While many traditional estimation methods rely on linear models, vehicle motion inherently exhibits complex, non-linear dynamics. Previous attempts to address this often resorted to piecewise linearization, introducing separate models for specific maneuvers (Yaakov Bar-Shalom, 2002). These methods then employed Interacting Multiple Model (IMM) filters, like IMM Kalman Filters, for state estimation (Park, 2022*a*,*b*). However, adopted linearized approaches suffer from several limitations described as follows:

- Lack of stability proofs: These linearization-based methods often lack formal guarantees of stability, raising concerns about their performance under real-world driving conditions.
- Limited maneuver coverage: The chosen linear models might not encompass the full spectrum of possible vehicle maneuvers, potentially leading to inaccurate tracking during more complex movements.
- Computational Demands: Implementing IMM filters, in particular, can be computationally expensive, requiring

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real-time evaluation of probabilities associated with each individual model, which can strain resources on embedded systems.

Therefore, it becomes crucial to explore alternative approaches that can effectively handle the inherent nonlinearities of vehicle motion without sacrificing stability, maneuverability, or computational efficiency.

Some previous works have explored the application of LMI-based nonlinear observers for vehicle state estimation (Jeon *et al.*, 2018, 2019) in the context of deterministic model-based estimators or observers. These nonlinear observers are derived from a single nonlinear model and have fixed observer gains, which makes them easy to implement. However, they also present some drawbacks, such as restricted stability regions and simplified model assumptions (e.g., constant velocity assumption). The nonlinear functions in the model limit the stability of the observers to a small range of steering angle and vehicle direction angle. To overcome this, switched gain observers (Rajamani *et al.*, 2020*a*,*b*) with different gains in different regions were needed to cover the whole operating range. Moreover, these observers also assumed constant velocity and cannot accurately cope with the estimation of the states of vehicles with varying velocities. In this paper we will design a modified version of a high-gain observer that exploits additional measurements, and hence, will be able to deal with variable velocities in addition to measurement noise, while guaranteeing stability and feasibility.

For nonlinear systems in the triangular form, the existence of stable high-gain observers is ensured if the nonlinear functions are Lipschitz (Gauthier and Kupka, 1994; Zemouche *et al.*, 2019; Alessandri, 2004; Alessandri and Rossi, 2015). Therefore, the high-gain observer always has a feasible solution, unlike the general class of LMI-based observers. Despite this advantage of high-gain observers, they have been rarely applied to real-world problems because converting nonlinear systems to the triangular form is challenging and not straightforward. Moreover, to the best of the authors' knowledge, the literature has scarcely used high-gain observers for multi-output systems. In this paper, we demonstrate how to design a high-gain observer for a vehicle system that has multiple outputs. The design method can be useful for other systems and may facilitate more practical applications of high-gain observers in the future.

Among the existing LMI-based methods, the LPV/LMI approach (Zemouche and Boutayeb, 2013) is the least conservative one. It is based on converting the dynamics of the estimation error into a polytopic system, and then applying the convexity principle to solve a finite number of LMI conditions without using large upper bounds to control the system's nonlinearity. For this reason, we use this method in this paper, combined with the standard high-gain observer technique, and we demonstrate that the LMIs can enhance the performance of the state observer.

Building upon our previous work in Bessafa *et al.* (2023), this paper introduces a more generalized high-gain observer framework. The main contribution lies in extending the classical high-gain observer methodology to accommodate multiple outputs, including nonlinear and constraint-based measurements, which are particularly relevant for vehicle state estimation and similar applications. The proposed observer incorporates additional sensor measurements beyond the standard position outputs (which makes it suitable for use in sensor fusion frameworks) and explicitly accounts for measurement noise. A novel set of Linear Matrix Inequality (LMI) conditions is derived, governed by a tunable parameter. An explicit constraint on this parameter is introduced to guarantee the solvability of the LMI and to ensure an Input-to-State Stability (ISS) bound on the estimation error. This design significantly enhances the observer's robustness and applicability in real-world scenarios characterized by noisy and intermittent measurements. In addition to validation performed on offline measurement data from the KITTI dataset, a detailed comparative study is conducted with both the Extended Kalman Filter (EKF) and the Standard High-Gain Observer (SHGO). While the EKF serves as a widely used benchmark, the comparison with SHGO using only position measurements clearly illustrates the benefit of integrating extra sensor outputs. The Multi-Output High-Gain Observer demonstrates superior performance, particularly under challenging conditions such as sensor noise and GPS data dropout. These results underscore the practical value of leveraging redundant measurements to improve estimation accuracy and robustness, marking a significant advancement in state estimation techniques for autonomous vehicle applications.

The paper is organized as follow: In Section 2, we revisit the standard high-gain observer and define the notation and preliminary tools used in the paper. Section 3 presents the vehicle kinematic model and its transformation into a triangular form. In Section 4, we introduce the new high-gain observer design methodology and provide proofs of stability. Section 5 demonstrates the performance of the proposed observer through various simulation and experimental scenarios, and comparisons are made with other methods. Finally, Section 6 concludes the paper with a summary of the findings and contributions.

2 Background and Preliminaries

This section is devoted to some useful preliminaries and background results.

2.1 Notation and preliminary tools

2.1.1 Notation

Throughout this paper, we use the following notation:

- $A^\top$ represents the transposed matrix of A;

- $\mathbb{I}_r$ represents the identity matrix of dimension r;
- for a square matrix S, $S>0$ ($S<0$) means that this matrix is positive definite (negative definite);
- $e_s(i) = \big(\underbrace{0,...,0,\overbrace{1}^{i\text{ th}},0,...,0}_{s\text{ components}}\big)^\top \in \mathbb{R}^s, s \geq 1$ is a vector of the canonical basis of $\mathbb{R}^s$;
- Consider two vectors $X = \begin{pmatrix} x_1 & \dots & x_n \end{pmatrix}^\top \in \mathbb{R}^n$ and $Y = \begin{pmatrix} y_1 & \dots & y_n \end{pmatrix}^\top \in \mathbb{R}^n$. For all $i = 0,...,n$, we define the auxiliary vector $X^{Y_i} \in \mathbb{R}^n$ corresponding to X and Y as $X^{Y_i} = \begin{pmatrix} y_1 & \dots & y_i & x_{i+1} & \dots & x_n \end{pmatrix}$ for $i = 1,...,n$ with the convention $X^{Y_0} = X$.

2.1.2 Preliminary tools

Lemma 1 (Zemouche and Boutayeb (2013)) *Consider a function $\Psi : \mathbb{R}^n \longrightarrow \mathbb{R}^p$. Then, the two following items are equivalent:*

- *Ψ is γ_Ψ-Lipschitz with respect to its argument, i.e.:*

$$\Big\|\Psi(X) - \Psi(Y)\Big\| \leq \gamma_\Psi \Big\|X - Y\Big\|, \quad \forall\, X, Y \in \mathbb{R}^n. \tag{1}$$

- *for all $i = 1,\dots,p$, $j = 1,\dots,n$, there exist functions*

$$\psi_{ij} \;:\; \mathbb{R}^p \times \mathbb{R}^n \longrightarrow \mathbb{R}.$$

and constants $\underline{\gamma}_{\psi_{ij}}$ and $\bar{\gamma}_{\psi_{ij}}$, so that $\forall\, X, Y \in \mathbb{R}^n$,

$$\Psi(X) - \Psi(Y) = \sum_{i=1}^{i=p} \sum_{j=1}^{j=n} \psi_{ij} \mathcal{H}_{ij}^{p,n} \Big(X - Y\Big), \tag{2}$$

and

$$-\gamma_\Psi \leq \underline{\gamma}_{\psi_{ij}} \leq \psi_{ij} \leq \bar{\gamma}_{\psi_{ij}} \leq \gamma_\Psi, \tag{3}$$

where

$$\psi_{ij} \triangleq \psi_{ij}\Big(X^{Y_{j-1}}, X^{Y_j}\Big) \;\; \text{and} \;\; \mathcal{H}_{ij}^{p,n} = \mathbf{e}_p(i)\mathbf{e}_n^\top(j).$$

2.2 Callback on the standard high-gain observer

This section is devoted to recall the standard high-gain observer methodology. In the standard high-gain observer we consider the following particular class of triangular systems with only a nonlinear term in the last component of the system:

$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \\ f(x) \end{bmatrix}, \\ y = x_1. \end{cases} \tag{4}$$

where $x \in \mathbb{R}^n$ is the state vector, $y \in \mathbb{R}$ is the measured output, and the function $f : \mathbb{R}^n \to \mathbb{R}$ satisfies the Lipschitz property formulated under the following form:

$$\left| f\left(x_1', \dots, x_n'\right) - f\left(x_1, \dots, x_n\right) \right| \leq \gamma_f \sum_{j=1}^{n} \left| x_j' - x_j \right| \tag{5}$$

for any x' and x in $\mathbb{R}^n$, where $\gamma_f \geq 0$ is the Lipschitz constant.

For the sake of brevity, we write system (4) under the form:

$$\begin{cases} \dot{x} = Ax + Bf(x), \\ y = Cx, \end{cases} \tag{6}$$

where

$$B = \begin{bmatrix} 0 & \dots & 0 & 1 \end{bmatrix}^\top, \quad C = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix}, \tag{7}$$

and the state matrix A is defined by

$$(A)_{i,j} = \begin{cases} 1 & \text{if} \quad j = i+1, \\ 0 & \text{if} \quad j \neq i+1. \end{cases} \tag{8}$$

The standard high-gain methodology considers the following state observer:

$$\dot{\hat{x}} = A\hat{x} + Bf(\hat{x}) + L(y - C\hat{x}), \tag{9}$$

with

$$L \triangleq T(\theta)K, \quad \theta > 1, \tag{10}$$

where

$$\mathrm{T}(\theta) \triangleq \operatorname{diag}\left(\theta^1, \dots, \theta^n\right) \text{ and } K \in \mathbb{R}^{n \times p}.$$

The following theorem is used in the high-gain methodology to design the observer gain K.

Theorem 2 (Gauthier and Kupka (1994)) *If there exist $P > 0$, $\lambda > 0$, Y and $\theta \geq 1$ such that*

$$A^\top P + PA - C^\top Y - Y^\top C + \lambda I < 0, \tag{11}$$

$$\theta > \theta_0 = \frac{2k_f \lambda_{\max}(P)}{\lambda}, \tag{12}$$

then for $K = P^{-1}Y^\top$, the observer (9) converges exponentially, where $k_f \geq \gamma_f$ is a constant independent from θ.

Proof: For more details on the proof, we refer the reader to Gauthier and Kupka (1994). The proof uses the Lyapunov function $\vartheta(\hat{e}) = \hat{e}^\top P \hat{e}$, where $\hat{e}$ is the transformed error defined as

$$\hat{e} \triangleq \mathrm{T}^{-1}(\theta)\big(x - \hat{x}\big).$$

Equation (11) is guaranteed to have a feasible solution if the pair (A,C) is observable, the observability being guaranteed for A and C from (7) and (8). $\square$

The high-gain observer theory extends to block-triangular letting us consider a nonlinear system with p outputs, each of relative degree r, and a nonlinear function $f(z) = [f_1(z), \ldots, f_p(z)]^\top \in \mathbb{R}^p$. After transformation into block-triangular form, the system becomes:

$$\dot{z} = \tilde{A}z + \tilde{B}f(z), \quad y = \tilde{C}z \tag{13}$$

where

$$\tilde{A} = I_p \otimes A_r, \quad A_r = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{r \times r} \tag{14}$$

$$\tilde{B} = \mathrm{diag}(B_1, \ldots, B_p), \quad B_i = [0 \ \cdots \ 0 \ 1]^\top \in \mathbb{R}^r \tag{15}$$

$$\tilde{C} = I_p \otimes C, \quad C = [1 \ 0 \ \cdots \ 0] \in \mathbb{R}^{1 \times r} \tag{16}$$

The high-gain observer is:

$$\dot{\hat{z}} = \tilde{A}\hat{z} + \tilde{B}f(\hat{z}) + L(y - \tilde{C}\hat{z}) \tag{17}$$

with

$$L = \mathrm{diag}(L_1, \ldots, L_p), \quad L_i = \theta^r \begin{bmatrix} l_1 \\ \theta l_2 \\ \vdots \\ \theta^{r-1} l_r \end{bmatrix} \tag{18}$$

Define the estimation error $\tilde{z} = z - \hat{z}$, and scaled error $e = T^{-1}(\theta)\tilde{z}$, where

$$T(\theta) = I_p \otimes \mathrm{diag}(\theta, \theta^2, \ldots, \theta^r) \tag{19}$$

Then the scaled error dynamics become:

$$\dot{e} = \theta(\tilde{A} - K\tilde{C})e + T^{-1}(\theta)\tilde{B}(f(z) - f(\hat{z})), K = T^{-1}(\theta)L \tag{20}$$

Assuming each f_i is Lipschitz with constant γ_i, i.e.,

$$|f_i(z) - f_i(\hat{z})| \leq \gamma_i \|\tilde{z}\|_2 \tag{21}$$

we obtain:

$$\|T^{-1}(\theta)\tilde{B}(f(z) - f(\hat{z}))\|_2^2 \leq \frac{1}{\theta^{2r}} \sum_{i=1}^{p} \gamma_i^2 \|\tilde{z}\|_2^2 \tag{22}$$

Using $\|\tilde{z}\|_2 = \|T(\theta)e\|_2 \leq \theta^r \|e\|_2$, we get:

$$\|T^{-1}(\theta)\tilde{B}(f(z) - f(\hat{z}))\|_2 \leq \sqrt{\sum_{i=1}^{p} \gamma_i^2} \cdot \|e\|_2 \tag{23}$$

Therefore, the nonlinearity is bounded by:

$$\|T^{-1}(\theta)\tilde{B}(f(z) - f(\hat{z}))\|_2 \leq k_f \|e\|_2, \quad k_f = \sqrt{\sum_{i=1}^{p} \gamma_i^2} \tag{24}$$

And the stability condition for the observer becomes:

$$\theta > \theta_0 = \frac{2k_f \lambda_{\max}(P)}{\lambda} \tag{25}$$

3 Ego Vehicle Model

In this section, we study the vehicle motion model and how to transform it into the triangular form required for the high-gain observer design.
Most real-world systems are not in the triangular form initially, so one needs to apply a coordinate transformation to convert the system into this form. The original vehicle model has many drawbacks. However, a modified model is much more efficient for transforming into the triangular form.

3.1 Vehicle Original Model

In this subsection, the triangular transformation of the vehicle model shown in Fig. 1 is investigated (Rajamani, 2012).

The model equations are described as follows:

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{\phi} \\ \dot{\delta}_f \end{bmatrix} = \begin{bmatrix} v\cos(\phi + \beta) \\ v\sin(\phi + \beta) \\ \frac{v}{l_f + l_r}\cos(\beta)\tan\left(\delta_f\right) \\ 0 \end{bmatrix}. \tag{26}$$

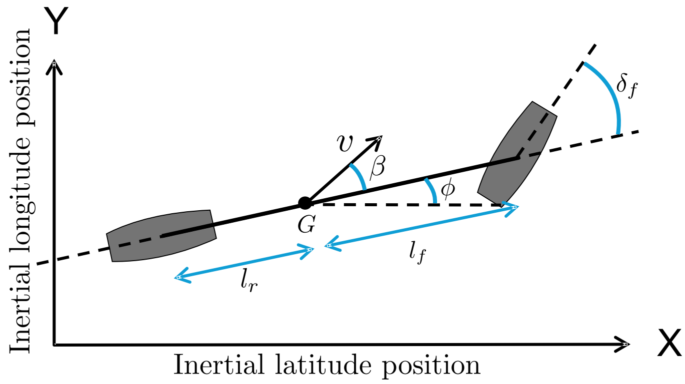


Fig. 1. Vehicle kinematic model.

where

- X relative longitudinal position of the vehicle.
- Y relative lateral position of the vehicle.
- ϕ yaw angle of the vehicle.
- δ_f steering angle of the front wheel.
- v is the total velocity
- β is the vehicle slip angle.
- l_f and l_r are the distances to the front and rear tires from the center of gravity of the vehicle.

We assumed that the derivative of the steering angle is zero. and the vehicle side slip is defined as follow :

$$\beta = \tan^{-1}\left(\frac{l_r \tan\left(\delta_f\right)}{l_f + l_r}\right) \tag{27}$$

and its derivative to be zero $\dot{\beta} = 0$.
The location of the vehicle is assumed to be roughly measured using RTK-corrected GPS. Therefore, the output equation is written as

$$y(t) = \begin{bmatrix} X(t) \\ Y(t) \end{bmatrix} + \omega(t). \tag{28}$$

where $\omega(t) \in \mathbb{R}^{2\times 1}$ is an additive measurement noise assumed to be uniformly bounded in magnitude as follows:

$$\exists\, \bar{\omega} \geq 0 \quad \text{such that} \quad \|\omega(t)\| \leq \bar{\omega}, \quad \forall t \geq 0.$$

3.2 *Triangular transformation*

The vehicle kinematic model poses challenges for nonlinear observer design due to its non-monotonic dynamics, which also complicates the direct application of a high-gain observer. To address these issues, some works in the literature (Rajamani *et al.*, 2020*a*; Jeon *et al.*, 2019) have proposed solutions using suitable transformations and a switched-gain based observer. However, a limitation of the switched-gain technique proposed in Rajamani *et al.* (2020*a*) is that the switching between regions of monotonicity is not systematic. To address this issue, we introduce a transformation that allows the application of a high-gain observer while preserving the original system constraints.
In order to obtain the model in the triangular form, each of the outputs X and Y need to be differentiated three times. By doing this, we get the following nonlinear transformation on the vehicle model (26):

$$\begin{cases} z_1 \triangleq X = y_1 \\ z_2 \triangleq \dot{X} = \dot{z}_1 = v\cos(\phi+\beta) \\ z_3 \triangleq \ddot{X} = \dot{z}_2 \\ z_4 \triangleq Y = y_2 \\ z_5 \triangleq \dot{Y} = \dot{z}_4 = v\sin(\phi+\beta) \\ z_6 \triangleq \ddot{Y} = \dot{z}_5 \end{cases} \tag{29}$$

We now derive the dynamical model resulting from the transformation (29). By construction of the transformation (29), the dynamics $\dot{z}_1$, $\dot{z}_2$, $\dot{z}_4$, and $\dot{z}_5$ are already known. Hence, it remains only to compute $\dot{z}_3$ and $\dot{z}_6$. Let $\theta = \phi + \beta$. Using the expressions $z_2 = v\cos\theta$ and $z_5 = v\sin\theta$ from (29), differentiation yields

$$\begin{aligned} z_3 &= \dot{z}_2 = -v\sin\theta\,\dot{\theta}, \\ z_6 &= \dot{z}_5 = \ \ v\cos\theta\,\dot{\theta}. \end{aligned} \tag{30}$$

These relations imply

$$\begin{aligned} -z_5z_3 + z_2z_6 &= -(v\sin\theta)\big(-v\sin\theta\,\dot{\theta}\big) + (v\cos\theta)\big(v\cos\theta\,\dot{\theta}\big) \\ &= v^2\sin^2\theta\,\dot{\theta} + v^2\cos^2\theta\,\dot{\theta} \\ &= v^2\,\dot{\theta}. \end{aligned} \tag{31}$$

Under the assumption $\dot{\beta} = 0$, we have $\dot{\theta} = \dot{\phi}$. Using (31) and $v^2 = z_2^2 + z_5^2$ (from (29)), we obtain

$$\dot{\phi} = \gamma(z) \triangleq \frac{-z_5z_3 + z_2z_6}{z_2^2 + z_5^2}, \tag{32}$$

where $z = \begin{bmatrix} z_1 & z_2 & z_3 & z_4 & z_5 & z_6 \end{bmatrix}^\top$ denotes the transformed state.

Note that $\gamma(\cdot)$ is a state-dependent function. It depends on the system state, particularly through the components of z. Moreover, by exploiting (26) together with (30) and $\ddot{\phi} = 0$, after expanding the corresponding expressions, we obtain

$$\dot{z}_3 = -\gamma(z)^2\, z_2, \tag{33}$$
$$\dot{z}_6 = -\gamma(z)^2\, z_5. \tag{34}$$

Therefore, the system in triangular form is written as follows:

$$\begin{cases} \dot{z}_1 = z_2, \\ \dot{z}_2 = z_3, \\ \dot{z}_3 = -\gamma(z)^2 \, z_2, \\ \dot{z}_4 = z_5, \\ \dot{z}_5 = z_6, \\ \dot{z}_6 = -\gamma(z)^2 \, z_5. \end{cases} \tag{35}$$

For more details on this transformation, we refer the reader to Alai *et al.* (2024).

For compactness we can write the system in (35) as:

$$\begin{cases} \dot{z} = \tilde{A}z + \tilde{B}f(z), \\ y = \tilde{C}z + \omega, \end{cases} \tag{36}$$

where

$$\tilde{A} = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}, \text{ with } A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\tilde{C} = \text{diag}\,(C, C), \text{ with } C = [1, 0, 0],$$

$$\tilde{B} = \text{diag}\,(B, B), \text{ with } B = [0, 0, 1]^\top,$$

and

$$f(z) = \begin{bmatrix} f_1(z) \\ f_2(z) \end{bmatrix} = \begin{bmatrix} -\gamma(z)^2 \, z_2 \\ -\gamma(z)^2 \, z_5 \end{bmatrix}. \tag{37}$$

Since the transformed model does not have the yaw angle as a state, we can calculate the value of ϕ using this equation:

$$\phi(t) = \tan^{-1}\left(\frac{z_5(t)}{z_2(t)}\right) - \beta. \tag{38}$$

The original system had four states while the system in the triangular form has six states. But there are two constraints among the six states.

Additional geometric constraints, derived from the original vehicle model through trigonometric relationships and modeling assumptions, can be incorporated into the new transformed model. These constraints ensure that key geometric properties of the original system, such as the non-holonomic behavior and the fixed wheelbase, are preserved within the transformed representation.

$$\dot{X}^2 + \dot{Y}^2 = v^2 \Rightarrow z_2^2 + z_5^2 = v^2, \tag{39}$$

$$\dot{X}\ddot{X} + \dot{Y}\ddot{Y} = 0 \Rightarrow z_2 z_3 + z_5 z_6 = 0. \tag{40}$$

Taking this constraint as an extra output, we have

$$h(z) = \begin{bmatrix} \sqrt{z_2^2 + z_5^2} \\ z_2 z_3 + z_5 z_6 \end{bmatrix} = \begin{bmatrix} v \\ 0 \end{bmatrix}.$$

Remark 3 *We acknowledge that the resulting nonlinear function in triangular form may not be globally Lipschitz. However, by restricting the system states to a bounded operating region, an assumption common in observer design for physical systems like vehicles, the Lipschitz condition holds locally. This ensures the convergence of the high-gain observer. In practice, the model admits an invariant compact set where nonlinearities are Lipschitz, allowing the use of Lipschitz extension techniques for observer design Zemouche and Rajamani (2022).*

4 New Multi-Output high-gain observer Design

This section is devoted to the primary theoretical advancements of the paper. Specifically, we introduce a novel generalized high-gain observer with multi-outputs capable of accommodating systems with extra measurements that deviate from the triangular form. We provide general observer design conditions, comprising a combination of LMI conditions and a threshold constraint on the high-gain parameter.

4.1 Motivation and objectives

The standard high-gain observer, while effective for certain applications, encounters limitations that necessitate its extension for broader usability. One critical restriction lies in its applicability only to systems in triangular form, rendering it unsuitable for a wide range of diverse system structures. Additionally, the standard high-gain observer faces challenges when dealing with multiple-output systems. Our proposed method addresses these limitations by providing a generalized framework that overcomes the constraints associated with the standard high-gain observer. Our approach ensures more versatile and robust application across various types of systems, enhancing the observer's practicality and effectiveness in real-world settings.

Remark 4 *The theory developed below is presented for a triangular system structure. However, as shown earlier in the context of high-gain observer design, the results and methodology can be similarly generalized to systems with multiple outputs organized in a block-triangular form. The same principles apply when extending the analysis to a general multi-block structure, provided each block corresponds to an output channel with a well-defined relative degree.*

Let us consider a second output as an additional measurement of the system, given by:

$$z = h(x)$$

With this, we can introduce the following generalized high-gain observer structure:

$$\dot{\hat{x}} = A\hat{x} + Bf(\hat{x}) + L(y - C\hat{x}) + \mathcal{M}(z - h(\hat{x})). \tag{41}$$

where L and M are two observer gains related to the measured outputs y and z respectively. As we exploit the high-gain methodology, the gain L is expressed under the form

$$L \triangleq \mathrm{T}(\theta)K,\ \mathcal{M} \triangleq \mathrm{T}(\theta)\mathcal{N}_\theta,\ \theta > 1, \tag{42}$$

with

$$\mathrm{T}(\theta) \triangleq \mathrm{diag}\left(\theta^1, \ldots, \theta^n\right),\ K \in \mathbb{R}^{n\times 1},\ \mathcal{N}_\theta \in \mathbb{R}^{n\times 1}.$$

and θ is the tuning high-gain parameter. The objective consists in determining the matrices K and $\mathcal{N}_\theta$, and the tuning parameter θ such that the estimation error $\tilde{e} = x - \hat{x}$ satisfies an exponential ISS bound to be established. The dynamics of the estimation error $\tilde{e}$ is given as follows:

$$\begin{aligned}\dot{\tilde{e}} = (A - LC)\tilde{e} + B[f(x) - f(\hat{x})] \\ - \mathcal{M}[h(x) - h(\hat{x})] - L\omega.\end{aligned} \tag{43}$$

Furthermore, before tackling the stability analysis, we introduce the transformed error

$$\varepsilon(t) \triangleq \mathrm{T}^{-1}(\theta)\tilde{e}(t). \tag{44}$$

The two nonlinear terms in (43), namely $\Delta f \triangleq f(x) - f(\hat{x})$ and $\Delta h \triangleq h(x) - h(\hat{x})$, will be handled in different ways. Indeed, Δf will be treated as in the standard high-gain methodology, while the term Δh will be transformed using the LPV/LMI technique in Zemouche and Boutayeb (2013). First, let us express the dynamics of the transformed estimation error, $\varepsilon(t)$, as follows:

$$\dot{\varepsilon} = \theta(A - KC)\varepsilon + \frac{1}{\theta^n}B\Delta f - \mathcal{N}_\theta \Delta h - K\omega. \tag{45}$$

By applying (Zemouche and Boutayeb, 2013, Lemma 7) on the Lipschitz function h, there exist functions $\psi_j : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$ and constants $\underline{\gamma}_j \le 0$ and $\bar{\gamma}_j \ge 0$, with $\underline{\gamma}_j \le \psi_j \le \bar{\gamma}_j$, such that

$$\Delta h = \left(\sum_{j=1}^{n} \theta^j \psi_j \mathrm{e}_n^\top(j)\right)\varepsilon. \tag{46}$$

Consequently, the dynamic equation of the transformed estimation error, ε, is expressed as follows:

$$\begin{aligned}\dot{\varepsilon} = \theta\left(A - \mathcal{N}_\theta\left(\sum_{j=1}^{n}\theta^{j-1}\psi_j\mathrm{e}_n^\top(j)\right) - KC\right)\varepsilon \\ + \frac{1}{\theta^n}B\Delta f - K\omega.\end{aligned} \tag{47}$$

On the other hand, since f is globally γ_f–Lipschitz, then there exists a constant $k_f \ge \gamma_f$, independent from θ, such that

$$\|\frac{1}{\theta^n}B\Delta f\| \le k_f\|\varepsilon\|. \tag{48}$$

It is quite clear that we cannot apply the standard high-gain methodology on the error system (47) due to the presence of the sum term depending on θ. Then, we need to handle this term by proposing a generalized high-gain based technique. This is the aim and content of the next section.

4.2 A generalized high-gain observer design method

This section is devoted to the main theoretical contribution of this paper. It aims to propose a novel observer design method generalizing the standard high-gain methodology only applicable for systems in triangular form. This novel technique consists in including an extra condition on the tuning parameter θ, which depends on the solvability of a set of LMI conditions.The presence of powers of θ makes it difficult to establish suitable conditions on θ that allow it to be chosen large enough while still ensuring an ISS bound on the estimation error. To address this challenge, we first assume the following:

$$\mathcal{N}_\theta = \frac{1}{\theta^n}\mathcal{N}, \tag{49}$$

where $\mathcal{N} \in \mathbb{R}^{n\times 1}$ is a constant matrix. By introducing the notation:

$$\mathcal{A}\left(\Psi^\theta\right) \triangleq -\sum_{j=1}^{n}\psi_j^\theta \mathrm{e}_n^\top(j), \tag{50}$$

where

$$\Psi^\theta \triangleq \begin{bmatrix}\psi_1^\theta & \ldots & \psi_n^\theta\end{bmatrix}^\top,\ \psi_j^\theta \triangleq \theta^{j-n-1}\psi_j,$$

it follows that the error system (47) can be simplified as follows:

$$\dot{\varepsilon} = \theta\Big(A + \mathcal{N}\mathcal{A}\left(\Psi^\theta\right) - KC\Big)\varepsilon + \frac{1}{\theta^n}B\Delta f - K\omega. \tag{51}$$

Since the functions $\psi_j, j = 1, \ldots, n$, are bounded, then the vector $\Psi \triangleq \begin{bmatrix}\psi_1 & \ldots & \psi_n\end{bmatrix}^\top$ belongs to a bounded convex set $\mathcal{S}$ for which the set of vertices is given by:

$$\mathcal{V} \triangleq \left\{v \in \mathbb{R}^n : v_j \in \{\underline{\gamma}_j, \bar{\gamma}_j\}\right\}. \tag{52}$$

Hence, the vector Ψ^θ belongs to a bounded convex set $\mathcal{S}^\theta$ for which the set of vertices is given by:

$$\mathcal{V}^\theta \triangleq \left\{v \in \mathbb{R}^n : v_j \in \{\underline{\gamma}_j\theta^{j-n-1}, \bar{\gamma}_j\theta^{j-n-1}\}\right\}. \tag{53}$$

The following inclusion is important for the sequel of the paper:

$$\forall\theta \ge 1, \sigma \ge 1,\ \theta \ge \sigma\ \Rightarrow\ \mathcal{S}^\theta \subseteq \mathcal{S}^\sigma. \tag{54}$$

Before presenting the result, we emphasize that the following theorem builds on the well-established high-gain observer framework while extending it to the multi-output LPV/LMI setting developed in this work. With this, we are now ready to summarize the design method in the theorem below.

Theorem 5 *Assume that there exist matrices $\mathcal{P}=\mathcal{P}^\top>0$, $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$, of appropriate dimensions, two scalars $\lambda>0$ and $\sigma\geq 1$, such that the following conditions are satisfied:*

$$\begin{bmatrix} \mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(v\big)\Big\}+\lambda\mathbb{I}_n & \mathcal{Y}^\top \\ \mathcal{Y} & -\mathcal{X}\end{bmatrix}<0, \quad \forall v\in\mathcal{V}^\sigma \tag{55}$$

$$\theta>\max\left(\sigma,\frac{2k_f\lambda_{\max}(\mathcal{P})}{\lambda}\right), \tag{56}$$

where $\mathcal{H}e\Big\{\mathbb{V}\Big\}\triangleq\mathbb{V}+\mathbb{V}^\top$ for any given matrix $\mathbb{V}$. Then, with $K=\mathcal{P}^{-1}\mathcal{Y}^\top$ and $\mathcal{N}=\mathcal{P}^{-1}\mathcal{Z}^\top$, the estimation error $\tilde{e}$ satisfies the following exponential ISS bound:

$$\begin{aligned}\|\tilde{e}(t)\|\leq\theta^{n-1}\sqrt{\frac{\lambda_{\max}(\mathcal{P})}{\lambda_{\min}(\mathcal{P})}}\|\tilde{e}_0\|\mathrm{e}^{-\frac{\kappa}{2}t} \\ +\frac{\theta^{n-\frac{1}{2}}}{\sqrt{\kappa}}\sqrt{\frac{\lambda_{\max}(\mathcal{X})}{\lambda_{\min}(\mathcal{P})}}\sup_{s\in[0,\ t]}\|\omega(s)\|,\end{aligned} \tag{57}$$

where

$$\kappa\triangleq\frac{\theta\lambda-2k_f\lambda_{\max}(\mathcal{P})}{\lambda_{\max}(\mathcal{P})}. \tag{58}$$

Proof: We use the quadratic Lyapunov function

$$\vartheta(\varepsilon)\triangleq\varepsilon^\top\mathcal{P}\varepsilon.$$

By introducing the change of variables $\mathcal{P}K=\mathcal{Y}^\top$, $\mathcal{P}\mathcal{N}=\mathcal{Z}^\top$, and after expanding the derivative of $\vartheta(\varepsilon)$ along the trajectories of (51), we obtain the following inequality:

$$\begin{aligned}\dot{\vartheta}(\varepsilon)\leq\theta\varepsilon^\top\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(\Psi^\theta\big)\Big\}\varepsilon \\ +2k_f\lambda_{\max}(\mathcal{P})\|\varepsilon\|^2+2\varepsilon^\top\mathcal{P}K\omega.\end{aligned} \tag{59}$$

By using the Young inequality (Zemouche *et al.*, 2017), we deduce that for any matrix $\mathcal{X}$ of appropriate dimensions, we have

$$\begin{aligned}\dot{\vartheta}(\varepsilon)&\leq\theta\varepsilon^\top\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(\Psi^\theta\big)\Big\}\varepsilon \\ &\quad+2k_f\lambda_{\max}(\mathcal{P})\|\varepsilon\|^2+\theta\varepsilon^\top\mathcal{Y}^\top\mathcal{X}^{-1}\mathcal{Y}\varepsilon+\frac{1}{\theta}\omega^\top\mathcal{X}\omega \\ &\leq\theta\varepsilon^\top\Bigg[\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(\xi\big)\Big\} \\ &\quad+\mathcal{Y}^\top\mathcal{X}^{-1}\mathcal{Y}\Bigg]\varepsilon+2k_f\lambda_{\max}(\mathcal{P})\|\varepsilon\|^2+\frac{\lambda_{\max}(\mathcal{X})}{\theta}\|\omega\|^2.\end{aligned}$$

On the other hand, from Schur Lemma, LMI (55) is equivalent to

$$\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(\Psi^\theta\big)\Big\}+\mathcal{Y}^\top\mathcal{X}^{-1}\mathcal{Y}\leq\lambda\mathbb{I}_n, \tag{60}$$

which means from the convexity principle and the inclusion (54) that for any $\theta\geq\sigma$, the inequality

$$\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(\Psi^\theta\big)\Big\}+\mathcal{Y}^\top\mathcal{X}^{-1}\mathcal{Y}\leq-\lambda\mathbb{I}_n, \tag{61}$$

is preserved. Hence, from (61), (56), and

$$\lambda_{\min}(\mathcal{P})\|\varepsilon\|^2\leq\vartheta(\varepsilon)\leq\lambda_{\max}(\mathcal{P})\|\varepsilon\|^2,$$

we can easily get

$$\dot{\vartheta}(\varepsilon)\leq-\kappa\vartheta(\varepsilon)+\frac{\lambda_{\max}(\mathcal{X})}{\theta}\|\omega\|^2, \tag{62}$$

which leads, by using the comparison theorem (Khalil, 1996), to

$$\begin{aligned}\|\varepsilon(t)\|\leq\sqrt{\frac{\lambda_{\max}(\mathcal{P})}{\lambda_{\min}(\mathcal{P})}}\|\varepsilon_0\|\mathrm{e}^{-\frac{\kappa}{2}t} \\ +\frac{1}{\sqrt{\theta}\sqrt{\kappa}}\sqrt{\frac{\lambda_{\max}(\mathcal{X})}{\lambda_{\min}(\mathcal{P})}}\sup_{s\in[0,\ t]}\|\omega(s)\|,\end{aligned} \tag{63}$$

Finally, by using the inequality

$$\theta\|\varepsilon(t)\|\leq\|\tilde{e}(t)\|\leq\theta^n\|\varepsilon(t)\|,\ \forall t\geq 0 \tag{64}$$

the ISS bound (57) is inferred. This completes the proof. □

Remark 6 *The LMI conditions* (55) *are always feasible. In other words, the LMIs* (55) *admit solutions for any function $h(.)$, which consistently ensures the design of our observer. This guaranteed feasibility stems from the parameterizations* (42) *and* (49) *of the matrices L, $\mathcal{M}$, and $\mathcal{N}_\theta$, respectively. Indeed, by applying Schur lemma, condition* (55) *is equivalent to*

$$\mathcal{H}e\Big\{\mathcal{P}A-\mathcal{Y}^\top C+\mathcal{Z}^\top\mathcal{A}\big(v\big)\Big\}+\mathcal{Y}^\top\mathcal{X}^{-1}\mathcal{Y}\leq\lambda\mathbb{I}_n. \tag{65}$$

Since the matrices A and C are in companion form, it follows that for any $\lambda > 0$, there exist matrices $\mathcal{P}$ and $\mathcal{Y}$ such that

$$\mathcal{H}e\Big\{\mathcal{P}A - \mathcal{Y}^\top C\Big\} < \lambda \mathbb{I}_n. \tag{66}$$

Therefore, one can always choose $\mathcal{Z}$ sufficiently small and $\mathcal{X}$ large enough to ensure that inequality (65) *is satisfied. Consequently, condition* (55) *also holds by Schur lemma.*

Remark 7 *Note that the same LMI conditions* (55) *also guarantee the $\mathcal{H}_\infty$ performance criterion. Specifically, from* (62)*, we have*

$$\dot{\vartheta}(\varepsilon(t)) + \kappa\vartheta(\varepsilon(t)) \leq \frac{\lambda_{\max}(\mathcal{X})}{\theta}\|\omega(t)\|^2. \tag{67}$$

By integrating both sides over time and using the fact that $\lim\limits_{t\to+\infty} \vartheta(\varepsilon(t)) \geq 0$, we obtain

$$\kappa\int_0^{+\infty} \vartheta(\varepsilon(s))\mathrm{d}s \leq \frac{\lambda_{\max}(\mathcal{X})}{\theta}\int_0^{+\infty}\|\omega(s)\|^2\mathrm{d}s + \vartheta(\varepsilon(0)). \tag{68}$$

Now, using the inequality

$$\lambda_{\min}\big(\mathcal{P}\big)\|\varepsilon(s)\|^2 \leq \vartheta(\varepsilon(s)) \leq \lambda_{\max}\big(\mathcal{P}\big)\|\varepsilon(s)\|^2$$

and the definition of the $\mathcal{L}_2$ norm

$$\|\xi\|_{\mathcal{L}_2} \triangleq \sqrt{\int_0^{+\infty}\|\xi(s)\|^2\mathrm{d}s}$$

for any vector ξ, and from (64)*, inequality* (68) *leads to the following $\mathcal{L}_2$ bound:*

$$\|\tilde{e}\|_{\mathcal{L}_2} \leq \frac{\theta^{n-\frac{1}{2}}}{\sqrt{\kappa}}\sqrt{\frac{\lambda_{\max}(\mathcal{X})}{\lambda_{\min}\big(\mathcal{P}\big)}\|\omega\|^2_{\mathcal{L}_2} + \frac{\lambda_{\max}(\mathcal{P})}{\lambda_{\min}\big(\mathcal{P}\big)}\|\tilde{e}(0)\|^2} \tag{69}$$

In the special case where $\tilde{e}(0) = 0$, this reduces to

$$\|\tilde{e}\|_{\mathcal{L}_2} \leq \frac{\theta^{n-\frac{1}{2}}}{\sqrt{\kappa}}\sqrt{\frac{\lambda_{\max}(\mathcal{X})}{\lambda_{\min}\big(\mathcal{P}\big)}}\|\omega\|_{\mathcal{L}_2}. \tag{70}$$

Inequality (70) *means that the $\mathcal{H}_\infty$ norm is satisfied as follows:*

$$\sup_{\omega\neq 0}\frac{\|\tilde{e}\|_{\mathcal{L}_2}}{\|\omega\|_{\mathcal{L}_2}} \leq \frac{\theta^{n-\frac{1}{2}}}{\sqrt{\kappa}}\sqrt{\frac{\lambda_{\max}(\mathcal{X})}{\lambda_{\min}\big(\mathcal{P}\big)}} \tag{71}$$

where the term on the right-hand side is referred to as the "disturbance attenuation level".
The ISS bound provides an instantaneous estimate of the error, showing exponential decay from the initial condition and bounded influence from a disturbance input in the supremum norm. In contrast, the $\mathcal{H}_\infty$ bound characterizes the worst-case energy gain from disturbance input to error output, using the norm $\mathcal{L}_2$.

Remark 8 *The ISS bound highlights how external disturbances, such as noise, affect the observer error dynamics and allows us to assess the robustness of the design. Two key parameters influence this behavior: the Lipschitz constant k_f and the tuning parameter θ. The Lipschitz constant k_f, which characterizes the nonlinearity of the system, is determined by the dynamics of the system and typically cannot be modified through design. In contrast, θ is a tunable parameter that directly affects the trade-off between noise attenuation and convergence speed. Specifically, decreasing θ improves noise attenuation by reducing the gain from noise to the estimation error, but this also slows down the convergence of the observer. In contrast, increasing θ speeds up convergence, but makes the observer more sensitive to noise. Therefore, careful tuning of θ is essential to balance robustness and performance. Furthermore, the term $\lambda_{\max}(\mathcal{X})$ in the ISS bound also influences noise sensitivity. To further enhance robustness, one can impose a constraint on $\lambda_{\max}(\mathcal{X})$ within the LMI formulation, thus controlling the magnitude of the observer gain and improving noise attenuation without significantly compromising convergence.*

Remark 9 *This paper assumes that the nonlinear function f is known and explicitly incorporated into the observer design. This assumption is made to clearly illustrate and validate the proposed methodology in a controlled setting. Nonetheless, the approach is general and can be extended to systems with partially or fully unknown nonlinearities. From a theoretical standpoint, handling such uncertainties requires only minor modifications to the Lyapunov-based analysis, consistent with standard high-gain observer techniques (e.g., using adaptive estimation or bounding functions for the unknown terms). Our focus here is on examining the effect of additional outputs such as extra measurements or implicit constraints that do not conform to the triangular structure-under known dynamics, to ensure clarity of exposition. The methodology remains broadly applicable and can be adapted to more complex scenarios, including uncertain dynamics, delayed measurements, and sampled outputs.*

5 Simulation and Experimental validation using the KITTI dataset

This section is devoted to the simulation results of the proposed observer design method applied to a vehicle model. Two different driving scenarios are analyzed. Experimental validation is performed using the KITTI dataset (via an offline processing of recorded data), with performance compared to both the standard high-gain observer and the EKF.

5.1 Observer design for the vehicle model

The process of designing our high-gain observer is initiated by transforming the vehicle model into triangular form, a crucial step to facilitate the subsequent observer design. Following this transformation, we proceed to solve the LMIs presented in Theorem 5, which provides us with the observer

gains K and $\mathcal{N}$.
For the vehicle model, we define the function $h(z)$ as:

$$h(z) = \begin{bmatrix} \sqrt{z_2^2 + z_5^2} \\ z_2 z_3 + z_5 z_6 \end{bmatrix},$$

where $z = (z_1, \ldots, z_6)^\top$ is the state vector.

The difference $\Delta h = h(X) - h(Y)$ can be expressed using a structured decomposition based on Lemma 1. Specifically, we write:

$$\Delta h = \sum_{i=1}^{2} \sum_{j=1}^{6} \psi_{i,j} H_{i,j}^{2,6} (Z - Y),$$

where

- $H_{2,6}^{i,j}$ is a 2×6 basis matrix with 1 at position (i, j) and 0 elsewhere,
- The coefficients $\psi_{i,j}$ are computed for each component $h_i(z)$ as follows.

For the coefficients $\psi_{i,j}$, they are computed as follows:

(1) **For** $h_1(z) = \sqrt{z_2^2 + z_5^2}$**:**

$$\psi_{1,1} = 0, \quad \psi_{1,3} = 0, \quad \psi_{1,4} = 0, \quad \psi_{1,6} = 0,$$
$$\psi_{1,2} = \frac{\xi_2}{\sqrt{\xi_2^2 + \xi_5^2}}, \quad \psi_{1,5} = \frac{\xi_5}{\sqrt{\xi_2^2 + \xi_5^2}},$$

where $\xi_2 \in (x_2, y_2)$ and $\xi_5 \in (x_5, y_5)$ are intermediate values.

(2) **For** $h_2(z) = z_2 z_3 + z_5 z_6$**:**

$$\psi_{2,1} = 0, \quad \psi_{2,4} = 0,$$
$$\psi_{2,2} = \xi_3, \quad \psi_{2,3} = \xi_2,$$
$$\psi_{2,5} = \xi_6, \quad \psi_{2,6} = \xi_5,$$

where ξ_k lies between x_k and y_k for $k \in \{2, 3, 5, 6\}$.

Using the coefficients $\psi_{i,j}$, we define a parameter-dependent matrix $A(\Psi^\sigma)$ as:

$$A(\Psi^\sigma) = -\sum_{i=1}^{2} \sum_{j=1}^{6} \psi_{i,j}^\sigma H_{i,j}^{2,6},$$

where the scaled coefficients $\psi_{i,j}^\sigma$ are given by:

$$\psi_{i,j}^\sigma \triangleq \sigma^{j-n-1} \psi_{i,j},$$

Concurrently, Lipschitz constants are computed, playing a vital role in determining the admissible values of the parameter θ.

The real system is then simulated using the model in (26), while the observer operates with the transformed system. It's noteworthy that speed measurements are incorporated as an additional output for the observer. This comprehensive design approach ensures the high-gain observer's effectiveness in accurately estimating the vehicle's trajectory, accounting for diverse factors and system dynamics.
The determination of the observer gain K and $\mathcal{N}$ involves solving the LMIs (55) for $\lambda = 10^7$, utilizing the SDPT-3 solver within the MATLAB software. As a constraint, we imposed that $Z > 10^4$ in order to have the eigenvalue of the gain $\mathcal{N}$ big enough in order to the added measurement has influence in our observer. The outcomes are as follows:

$$\mathcal{P} = \begin{bmatrix} 6.5 & -1.5 & -2.3 & 0 & 0 & 0 \\ -1.5 & 3.0 & -1.5 & 0 & 0 & 0 \\ -2.3 & -1.5 & 6.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6.5 & -1.5 & -2.3 \\ 0 & 0 & 0 & -1.5 & 3.0 & -1.5 \\ 0 & 0 & 0 & -2.3 & -1.5 & 6.5 \end{bmatrix},$$

$$K = \begin{bmatrix} 392.8 & 680.9 & 392.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 392.8 & 680.9 & 392.7 \end{bmatrix}^\top,$$

$$\mathcal{N} = \begin{bmatrix} 4.72 & 8.14 & 4.71 & 4.72 & 8.14 & 4.71 \\ 9.72 & 1.14 & 0.71 & 9.72 & 1.14 & 0.71 \end{bmatrix}^\top.$$

We compute the parameter k_f that depends on the Lipschitz constant of the nonlinear function f defined in (37). By computing the Jacobian and bounding the states we find $k_f = 0.1005$. With this value, we get $\theta_0 = 1.36$ and $\sigma = 5$. Then, we can choose $\theta = 5.2$.

5.2 *Simulation results*

We simulate two different scenarios, both with a constant speed; we take $l_f = 1.45$ and $l_r = 1.55$. In the first scenario, the vehicle travels in a straight line, and in the second, it makes a left turn. In both scenarios, bounded measurement noise is added to the position measurements. The noise samples are drawn from a Gaussian distribution with standard deviation $\sigma = 10^{-2}$ and truncated to the interval $[-3\sigma, 3\sigma]$ to satisfy the bounded-noise assumption used in the theoretical analysis.

5.2.1 *Scenario Driving Straight with Measurement Noise*

In this simulation scenario, we consider a vehicle moving along a straight line with a constant speed $v = 5m/s$ on the X axis. To simulate real-world conditions and assess the robustness of our observer model, we introduce noise into the position measurements. This noise is intended to mimic the inaccuracies that might occur in actual position tracking systems due to various factors such as sensor errors or environmental disturbances.

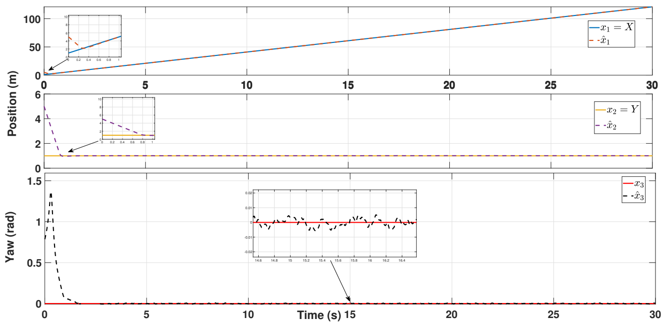


Fig. 2. Vehicle states estimation in straight driving scenario.

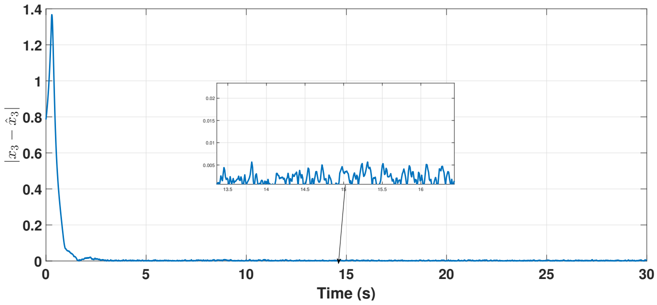


Fig. 3. Yaw angle estimation error in straight driving scenario.

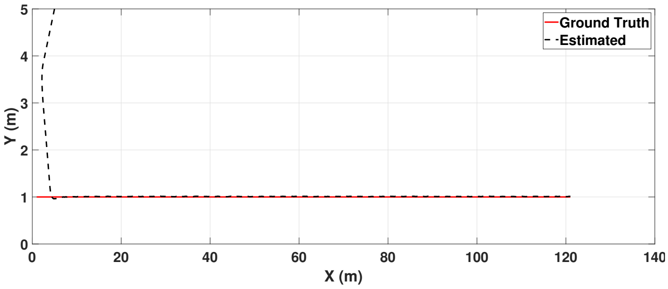


Fig. 4. Trajectory estimation in straight driving scenario.

5.2.2 *Scenario Turning Left*

In this simulation scenario, we examine a situation where the vehicle is executing a left turn. The steering angle is varied slowly to maintain the assumptions of our model. Throughout this maneuver, the vehicle maintains a constant speed, this scenario allows us to study the system's response to changes in steering input, providing valuable insights into the performance and stability of our observer model under varying driving conditions.

As illustrated in Figures 2 and 5, the observer estimates the yaw angle accurately and converges to the true values in both scenarios. The oscillation and the peaking phenomenon

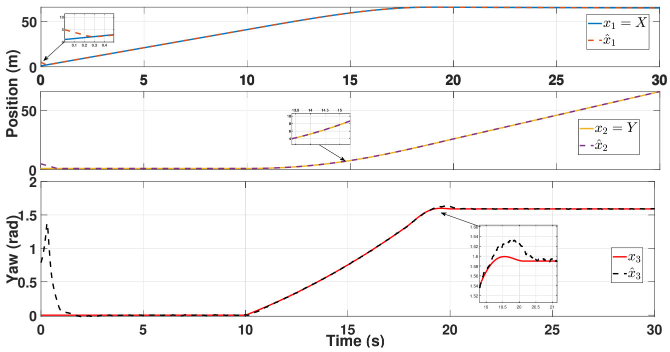


Fig. 5. Vehicle states estimation in left turn scenario.

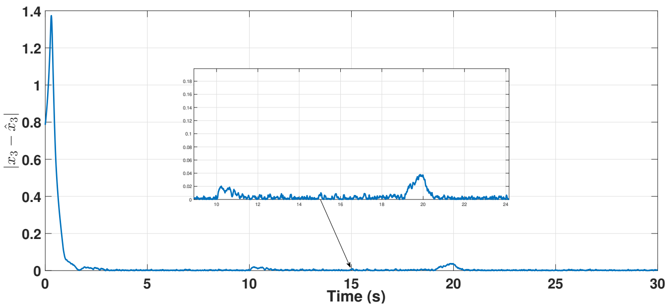


Fig. 6. Yaw estimation error in left turn scenario.

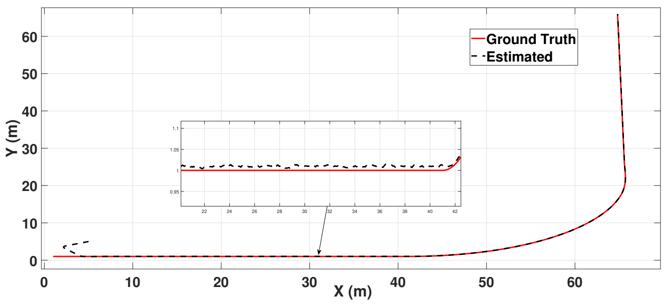


Fig. 7. Trajectory estimation in left turn scenario.

of the estimated signal, which are common challenges of the high-gain observer, are also minimized. These results validate the effectiveness of our proposed method. Moreover, the Figures 3 and 6 present the yaw angle estimation error and the quick convergence of the observer. The Figures 4 and 7 show the position estimation, which filters out the measurement noise and estimates the vehicle states.

5.3 *Experimental Results*

This section presents the experimental results obtained using an available dataset and includes comparisons with both the EKF and the standard high-gain observer.

5.3.1 *KITTI Dataset*

In addition to simulations, we conducted experimental evaluations to show the effectiveness of our observer design using the KITTI dataset Geiger *et al.* (2013). This dataset consists of data collected by a vehicle equipped with a 6-axis, 100 Hz GPS/IMU (OXTS RT3003) with L1/L2 RTK, and a resolution of $0.01m$ / $0.1°$, as it navigated through a city. In this context, our focus was on localizing the ego vehicle itself, not a target vehicle, using the same kinematic nonlinear model. By using an inertial GPS with Real-Time Kinematic (RTK) correction, we can obtain highly accurate position measurements to $1cm$ in position measurement that will be used as ground truth to compare with the estimated states. The dataset is acquired and synchronized at a rate of 10 Hz. We used the geographic coordinates (longitude and latitude), global orientation, and velocities. Consequently, we believe that the novel approach of incorporating additional outputs has the potential to significantly improve performance compared to the traditional high-gain observer in this specific application.

Fig. 8. Camera Capture from KITTI Dataset Geiger *et al.* (2013).

Upon processing the raw data obtained from the GPS sensor, we can extract the vehicle position and velocity. We utilize the filtered Inertial Measurement Unit (IMU) measurement of the yaw angle as the ground truth. This ground truth is then compared with our estimation for validation purposes.

In our experimentation setup, the scenario involves the car traversing a dynamic path, starting with a variable speed along a straight line, followed by a left turn, and then continuing along a curved right path. The vehicle eventually comes to a halt at a red light. The entire scenario spans 45 seconds, but we focused on a 35-second interval as our interest lies primarily in instances when the vehicle is in motion. To address potential short communication gaps in the GPS data, we leveraged interpolation in the KITTI dataset. Subsequently, we explore scenarios where there is data or signal loss, assessing the performance of our observer in estimating the vehicle trajectory under such conditions.

The experimental data confirm the effectiveness of our proposed observer. The Figure 9 shows that the observer estimates the vehicle yaw angles well and converges fast, as seen in the Figure 10. The position estimation from the GPS signal is precise, as shown in the Figure 11. The velocity estimation is also good, as shown in the Figure 12, which demonstrates the effectiveness of the methodology. Moreover, the observer can estimate the vehicle states even with varying velocity and steering angles, as shown in the figures, which is very powerful.

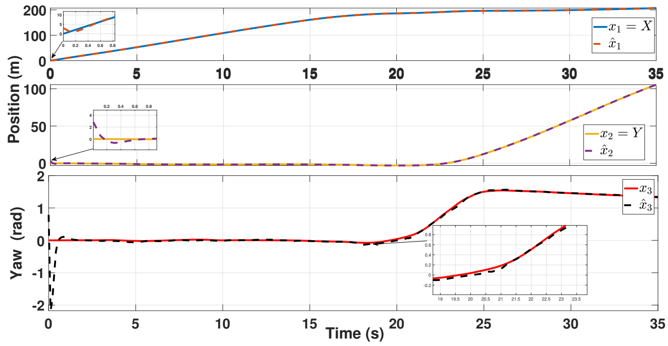


Fig. 9. States estimation using real data.

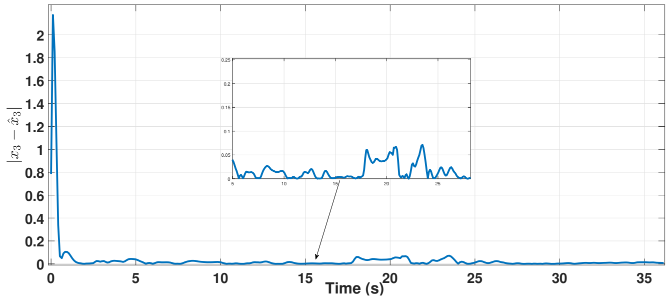


Fig. 10. Yaw estimation error.

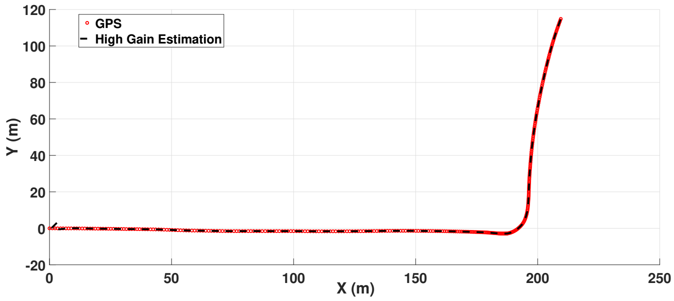


Fig. 11. Trajectory estimation.

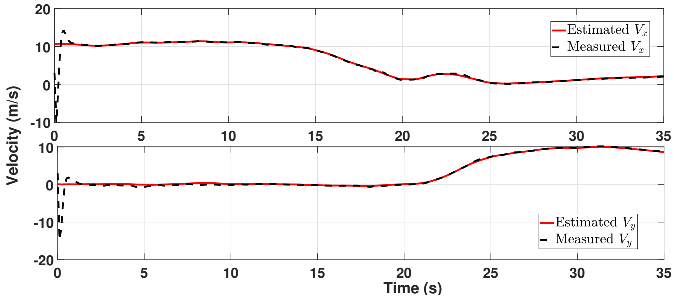
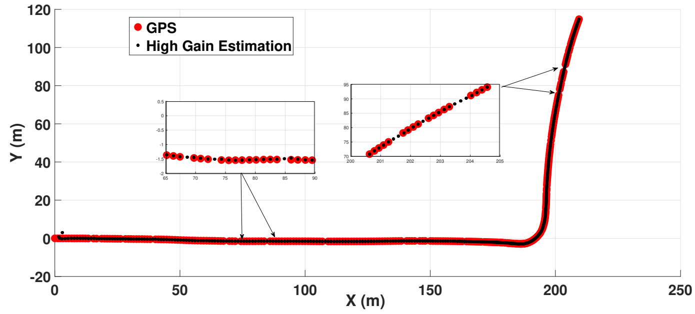
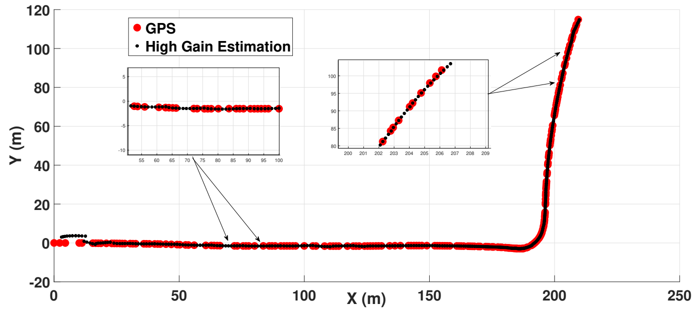


Fig. 12. Velocity estimation on both axis (V_x and V_y).

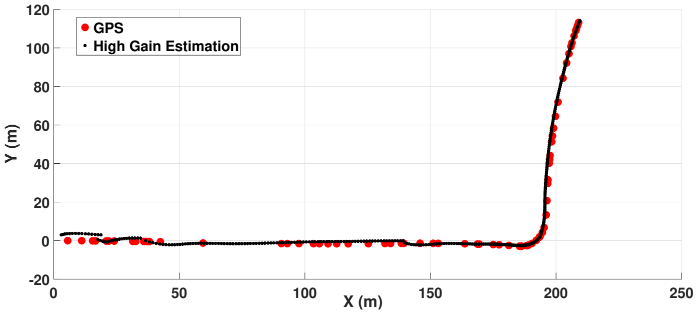
As said before, the GPS signal can have some short drop-outs. To show the effectiveness of our proposed method, we will simulate data drop-outs of 10%, 40%, and 80% by randomly removing the position information from the measurement vector Y. We will only have the velocity information of the vehicle and we will check if we can estimate the vehicle trajectory.



(a) 10% Missing Data.



(b) 40% Missing Data.



(c) 80% Missing Data.

Fig. 13. Simulation of data gaps in GPS signal.

The Fig. (13) shows that we can estimate the vehicle trajectory well only with the velocity information, especially for the cases of 10% and 40% drop-out. The error is slightly larger for the case of 80% drop-out, but it is acceptable considering that we have no position information and we use a simplified vehicle model. These figures demonstrate the robustness of the proposed observer.

5.3.2 *Comparison with the Extended Kalman Filter (EKF)*

To showcase the advantages of the high-gain observer, we conducted a performance comparison with a traditional estimation filter, the Extended Kalman Filter (EKF), under a specific scenario depicted in Figure 14. The EKF was designed based on model (26), which incorporates acceleration into the state vector. This comparison helps to highlight the unique advantages offered by the high-gain observer. The EKF was selected as a benchmark not only because of its widespread adoption in nonlinear state estimation but also due to its central role in sensor fusion for autonomous systems. Its widespread use in combining measurements from heterogeneous sensors makes it a relevant and practical reference point. Comparing our observer to the EKF under identical constraints and outputs allowed for a fair and meaningful evaluation of performance in realistic conditions. To ensure a fair comparison with the EKF, we utilized only the velocity v as an additional measurement for updating the gains. In addition, we strictly adhered to using only GPS measurements, specifically position and velocity, to maintain consistency in our approach. Moreover, to demonstrate the rate of convergence, we initialized both observers with identical initial conditions that were significantly different from the first estimation.

In this experiment, we will adopt a scenario where we intentionally create a drop-out of 40% :

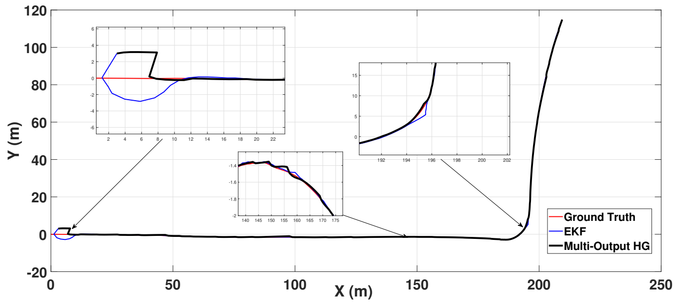


Fig. 14. Trajectory estimation.

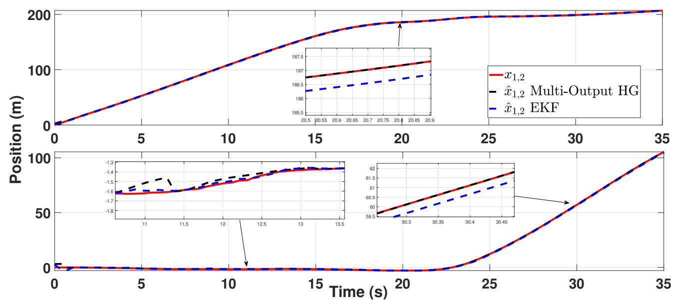


Fig. 15. State estimation (Position).

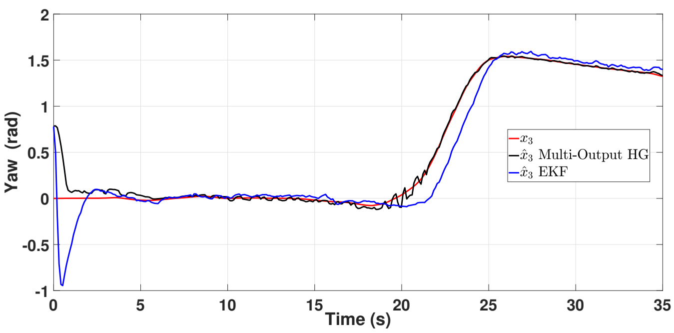


Fig. 16. Yaw angle estimation.

In the given scenario, where we experience a 40% loss in expected GPS position measurements, the high-gain observer outperforms the Extended Kalman Filter (EKF) in terms of convergence rate as anticipated as shown in Figures 15 and 14. The high-gain observer converges significantly faster than the EKF. Furthermore, the accuracy of the EKF is marginally inferior to that of the high-gain observer. This discrepancy can be attributed to the variance matrix of the EKF. Under the same assumptions of a null yaw rate, the estimation of the yaw angle in Figure 16 is intermittently more precise with the high-gain observer. However, it is important to note that the high-gain observer consistently exhibits peaking, especially when there are substantial changes in state.

In vehicle tracking, it is common to incorporate multiple measurements using sensor fusion, usually using an Extended Kalman Filter (EKF) (Leonard and Durrant-Whyte, 1991; von der Hardt *et al.*, 1996; Ullah *et al.*, 2021). In our case to achieve this, we can introduce an additional output or measurement, denoted as h_3. For instance, if we consider the integration of an Inertial Measurement Unit (IMU) to measure the yaw rate, it can be utilized as demonstrated in equation (72)

$$h_3(z) = \dot{\phi}(t) = \gamma = \frac{(-z_5 z_3 + z_2 z_6)}{z_2^2 + z_5^2}, \tag{72}$$

Thus, the additional measurement will now be expressed as follows:

$$h(z) = \begin{bmatrix} h_1(z) \\ h_2(z) \\ h_3(z) \end{bmatrix} = \begin{bmatrix} \sqrt{z_2^2 + z_5^2} \\ \\ z_2 z_3 + z_5 z_6 \\ \\ \frac{(-z_5 z_3 + z_2 z_6)}{z_2^2 + z_5^2} \end{bmatrix}. \tag{73}$$

In our study, we compare (EKF) with our observer. To emulate the inherent variance of a GPS, we introduce Gaussian noise to the GPS position measurements, characterized by a standard deviation (σ) of $1.5m$. This simulation aligns with our findings that revealed a significant uncertainty of approximately ± 1.5 meters in both X and Y dimensions when using a conventional and standard GPS without (RTK) correction. However, the velocities $\dot{X}$ and $\dot{Y}$ exhibited a higher degree of precision, with an uncertainty of about $\pm 0.05 m/s$, attributable to the carrier-phase signals.

By applying the same theory as in Theorem 5, we can determine the observer gains. This time, in LMI (55), we assign more significance to the extra-output. This means we constrain the matrix $\mathcal{N}$ to have a more substantial gain than matrix K. The result is as follows:

$$K = \begin{bmatrix} 0.2247 & 0.4704 & 0.1832 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2247 & 0.4704 & 0.1832 \end{bmatrix}^{\top},$$

$$\mathcal{N} = \begin{bmatrix} 0.4858 & 0.7402 & 0.4603 & 0.4858 & 0.7402 & 0.4603 \\ 0.8858 & 0.3402 & 0.1603 & 0.8858 & 0.3402 & 0.1603 \\ 0.4858 & 0.7402 & 0.4603 & 0.4858 & 0.7402 & 0.4603 \end{bmatrix}^{\top}.$$

and we take value of $\theta = 3$.

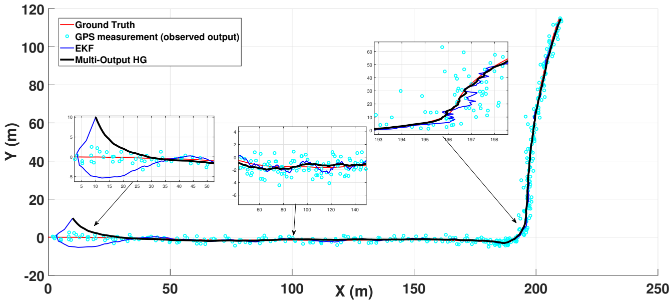


Fig. 17. Trajectory estimation.

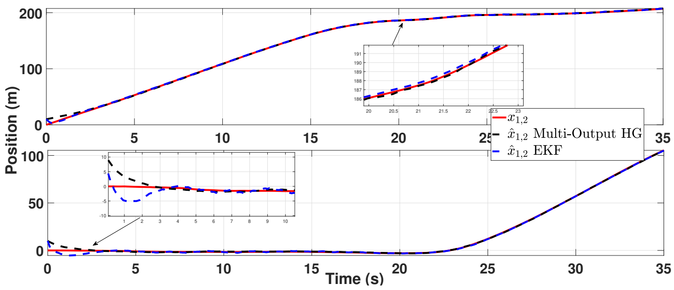


Fig. 18. State estimation (Position).

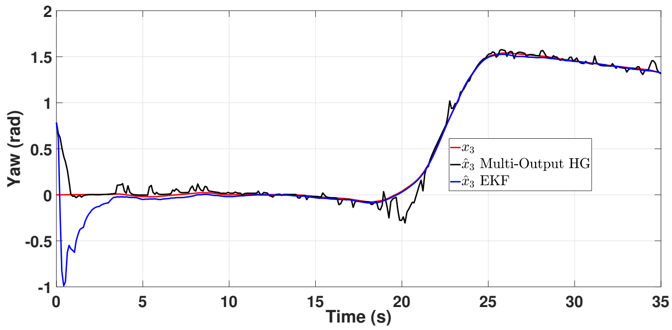


Fig. 19. Yaw angle estimation.

In this scenario, where measurement noise is introduced on the vehicle position, we have adjusted the gains and the parameter θ. As a result, the measured speed and yaw rate have a greater influence on the correction of the states. However, this adjustment leads to a slight decrease in the convergence rate. Additionally, the estimation of the yaw angle exhibits increased oscillation and noise, yet it remains accurate when changes are minimal. It's noteworthy that the Extended Kalman Filter (EKF) demonstrates superior performance in such scenarios as shown in Fig. (19, particularly when there are changes in the yaw rate. Nevertheless, the high-gain observer, still outperforms the EKF in position correction and estimation as shown in Fig. (17),(18). Furthermore, our method by employing the triangular model, we can extend our estimations to other states such as longitudinal and lateral accelerations and velocities.While the EKF can provide better noise attenuation, the proposed method ensures the stability of the estimation even in the presence of measurement noise.

5.3.3 Comparison with the standard high-gain observer

To highlight the improvements achieved by leveraging additional measurements, we compare the proposed multi-output high-gain observer with the Standard High-Gain Observer (SHGO), which utilizes only the position output as described in equation (28). In contrast, the multi-output observer incorporates three additional geometric and kinematic measurements as detailed in equation (73). This comparison is conducted to demonstrate how the inclusion of extra outputs enhances state estimation accuracy and robustness. Specifically, we examine the observer's performance under nominal conditions as well as during sensor degradation scenarios such as GPS signal dropout or failure-common occurrences in real-world applications.

The gain matrix K for the Standard High-Gain Observer (SHGO) was computed by solving the LMI condition (11) in Theorem 2. To ensure a fair comparison, we selected the same tuning parameter value $\theta = 3$ as used in the multi-output observer.
In the first scenario, the position measurements are affected by additive zero-mean Gaussian noise characterized by a standard deviation of $\sigma = 0.1$.

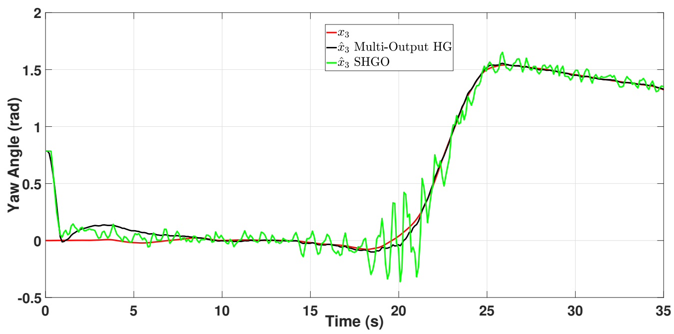


Fig. 20. Yaw angle estimation.

As illustrated in Fig. 20, the yaw angle estimation for both observers exhibits similar convergence times, which is expected given the identical tuning parameter value θ. However, the Multi-Output High-Gain Observer demonstrates superior performance in the presence of measurement noise, yielding more accurate and robust estimates.

We also conducted a comparative test under conditions of data dropout. In this scenario, measurement noise was not considered; instead, 40% of the position measurements were randomly removed to simulate data loss. As shown in Fig. 21 and Fig. 22 , the Standard High-Gain Observer (SHGO) struggles to estimate the states accurately, which is expected given that it operates effectively in open-loop during periods without measurement updates. In contrast, the Multi-Output High-Gain Observer, leveraging the additional measurements, maintains robust and accurate state estimation-demonstrating the value of these extra outputs in enhancing estimation resilience. It is also important to note that beyond 40% dropout, the SHGO tends to diverge, whereas the Multi-Output observer continues to provide acceptable estimates, as demonstrated in the previous subsection.

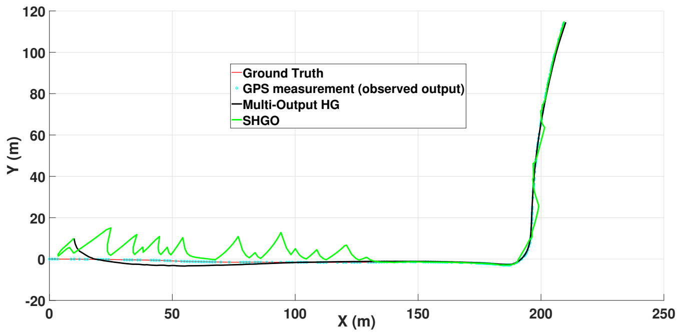


Fig. 21. Simulation of data gaps in GPS signal (40%) to compare with SHGO

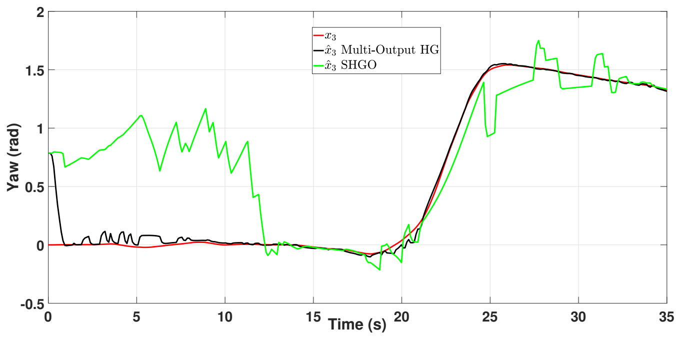


Fig. 22. Yaw angle estimation with 40% data gaps.

6 Conclusion

In this paper, the development of a multi-output high-gain observer has shown significant promise in tracking the trajectory of a vehicle. The proposed method ensures input-to-state stability while effectively handling multiple outputs, including additional nonlinear measurements. By transforming the vehicle's kinematic model into a triangular form, we enabled the application of high-gain observer techniques to a broader class of systems.

Simulations across various driving scenarios, along with experimental validation using the KITTI dataset, demonstrate the observer's ability to accurately localize the vehicle and estimate its orientation. In addition to a comparison with the Extended Kalman Filter (EKF), where our method showed notable improvements in both accuracy and convergence speed, we also conducted a detailed comparison with the Standard High-Gain Observer (SHGO). The results confirmed that the proposed multi-output observer provides enhanced estimation performance, particularly in the presence of measurement noise and during data dropout events, due to the integration of additional outputs. Furthermore, the structure of the proposed observer is well-suited for sensor fusion applications, expanding its potential for real-world deployment. Thus, this new high-gain observer methodology presents a promising avenue for future research and practical applications in vehicle trajectory tracking and state estimation. Future work will focus on applying observer-based control strategies to further enhance vehicle performance and robustness with respect to unknwon nonlinearities.

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